PD MODELING — METHODOLOGY & VALIDATION

Home Credit Default Risk

XGBoost benchmark and a fully auditable WoE/IV logistic regression scorecard

Technical deep dive: feature engineering, variable selection, multicollinearity diagnostics, and validation protocol

Kaggle: Home Credit Default Risk

SCOPE & VALIDATION PROTOCOL

Dataset, scope decision, and evaluation methodology

307,511	8.07%	156	80/20
clients (application_train + aggregated bureau.csv)	historical bad rate	columns after merge (154 candidate features)	stratified train/validation split, seed fixed

SCOPE DECISION

This slice of the pipeline is intentionally limited to application_train.csv plus aggregated bureau.csv, not the full relational schema (previous_application, POS_CASH_balance, credit_card_balance, installments_payments). The goal was to validate a complete pipeline end to end before scaling data complexity, a deliberate MVP-style sequencing rather than a scope gap.

VALIDATION METHODOLOGY

- XGBoost: stratified 5-fold CV for the performance estimate, plus a fixed 80/20 split for SHAP and gain-curve artifacts
- Scorecard: WoE tables and bin edges computed strictly on the training split, applied (never recalculated) on validation
- Gain curves use out-of-fold predictions (XGBoost, full base) or held-out validation (scorecard); the two are not on identical populations, noted explicitly wherever compared

Column-by-column aggregation of bureau.csv

bureau.csv (1,716,428 rows, 17 columns, 5.61 rows/client) aggregated to SK_ID_CURR level using an explicit per-column rule, not blanket mean/sum:

Rule	Applied to	Rationale
sum	AMT_CREDIT_SUM, AMT_CREDIT_SUM_DEBT, AMT_CREDIT_SUM_LIMIT, AMT_ANNUITY	Total exposure is additive across credits
mean	DAYS_CREDIT, CREDIT_DAY_OVERDUE, AMT_* fields	Typical per-credit profile, normalized by count
max	CREDIT_DAY_OVERDUE, AMT_CREDIT_MAX_OVERDUE, DAYS_CREDIT_UPDATE	Worst-case signal a mean would dilute
count / crosstab	CREDIT_ACTIVE (4 categories)	Low cardinality, equivalent to one-hot via count
nunique + mode count	CREDIT_TYPE (15 categories)	Diversification and concentration signals
excluded	CREDIT_CURRENCY	99.92% concentrated in one category, no discriminative power

Result: 29 features, bureau_agg (305,811 clients). Merged into application_train via SK_ID_CURR (left join): 44,020 clients (14.3%) have no bureau history, an MNAR pattern consistent with prior EDA findings, not a merge defect.

Hyperparameters and 5-fold stratified validation

HYPERPARAMETERS

n_estimators	200
max_depth	4
learning_rate	0.05
scale_pos_weight	11.39 (train class ratio)
tree_method	hist
enable_categorical	True (native pandas category dtype)

No tuning performed at this stage; near-default hyperparameters used as a reference point (see Limitations).

5-FOLD STRATIFIED CV (154 FEATURES)

Fold	AUC
1	0.7558
2	0.7640
3	0.7570
4	0.7628
5	0.7544
Mean (std)	**0.7588 (±0.0039)**

HOLDOUT

0.7614

single 80/20 split, within fold range

The holdout AUC (0.7614) falls inside the fold range (0.7544–0.7640), confirming it is not a favorable-split artifact. Standard deviation (0.0039) is small relative to the mean, indicating stable generalization.

Redundancy diagnostics in the 15-feature XGBoost variant

Top 15 selected by native feature importance from the train-only baseline model (avoiding validation-based selection bias). Two suspected redundancies were tested empirically, not assumed from correlation alone:

CONFIRMED: DAYS_EMPLOYED removed

DAYS_EMPLOYED and YEARS_EMPLOYED are a deterministic linear transform of each other (YEARS = -DAYS/365). Both appeared in the naive top-15.

Mean k-fold AUC: 0.7514 → 0.7525

Removing the exact duplicate improved AUC (AMT_CREDIT was promoted into the set), confirming zero information loss.

TESTED AND REVERTED: AMT_GOODS_PRICE kept

AMT_GOODS_PRICE correlates 0.99 with AMT_CREDIT, but is not a deterministic transform (down payments, partial financing cause divergence).

Mean k-fold AUC: 0.7525 → 0.7499 (removed)

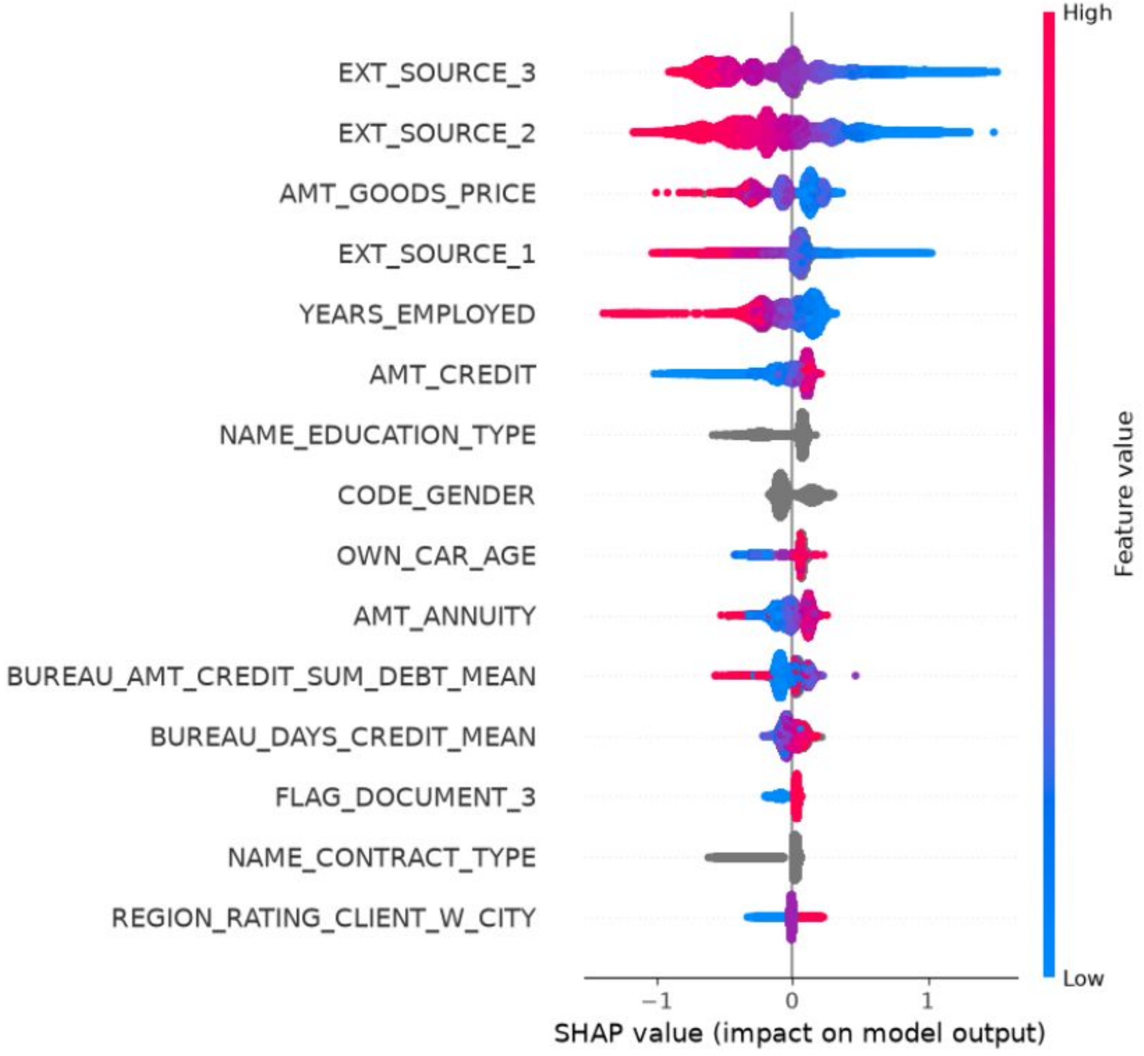
Removal measurably hurt AUC: high correlation without an exact functional relationship still carried marginal signal. Both variables kept.

Principle: correlation is a reason to test removal, not a rule to remove automatically. Final: 15 features, mean k-fold AUC 0.7525, holdout AUC 0.7538.

SHAP (aggregate)

SHAP — TOP DRIVERS (15-feature model)

- EXT_SOURCE_3, EXT_SOURCE_2, EXT_SOURCE_1: dominant, high value pushes prediction toward lower risk
- AMT_GOODS_PRICE, AMT_CREDIT, AMT_ANNUITY: higher values associate with lower risk
- Categorical variables (NAME_EDUCATION_TYPE, CODE_GENDER, NAME_CONTRACT_TYPE) show no color gradient: real effect, no "more/less" scale
- Limitation: explains the model in aggregate, not any individual applicant's score



Why WoE/IV, and the underlying formulas

RATIONALE

- XGBoost captures non-linearity natively (tree splits); WoE exists specifically to give a linear model that same capability
- Logistic regression's additive structure (intercept + Σ coefficient × WoE) is exactly what a point table requires
- Regulatory expectation in credit risk: an auditable, coefficient-level explanation for every decision

$$\mathrm{WoE} = \ln(\ \%good\ /\ \%bad\ )$$

$$\mathrm{IV} = \Sigma\ (\%good - \%bad) \times \mathrm{WoE}$$

%good/%bad computed with additive (Laplace) smoothing, α = 0.5, to avoid ln(x/0) when a bin has zero bad cases.

IV THRESHOLDS (SIDDIQI)

IV range	Strength
< 0.02	No discrimination
0.02 – 0.10	Weak
0.10 – 0.30	Medium
0.30 – 0.50	Strong
> 0.50	Suspicious (possible leakage)

Full IV scan, concept-level deduplication, instability correction

155 candidate variables scanned for IV. Concept-based deduplication (regex grouping of _AVG/_MODE/_MEDI variants and DAYS_X/YEARS_X pairs, keeping the highest-IV variable per group) reduced this to 121.

INSTABILITY CORRECTIONS APPLIED TO THE TOP-40 BUFFER (11 VARIABLES FLAGGED)

Group	Variables	Correction
1 — sparse Missing bin	EXT_SOURCE_2, AMT_GOODS_PRICE, DAYS_LAST_PHONE_CHANGE	Merged into nearest numeric bin by WoE
2 — single rare category	CODE_GENDER (XNA, 4 records)	Left as documented residual; insufficient volume for any merge criterion
3 — many small categories	OCCUPATION_TYPE, ORGANIZATION_TYPE, NAME_INCOME_TYPE, WALLSMATERIAL_MODE	Collapsed into 'Other_grouped', cascading merge into nearest large category if still unstable
4 — dominant Zero value	BUREAU_AMT_CREDIT_SUM_DEBT_SUM (29.2% zero), BUREAU_AMT_CREDIT_SUM_LIMIT_MEAN (79.9% zero)	Zero isolated as its own explicit bin

AMT_GOODS_PRICE required a dedicated resolution: naive 10-bin WoE was non-monotonic (not a small-sample artifact, all bins > 18,900 records). Finer binning (20 bins) raised IV from 0.0919 to 0.1025; iterative surgical merging (smallest WoE-difference pairs first) reduced to 12 bins at IV 0.1006, validated stable across 5 folds (WoE std 0.005–0.037 per bin). Later excluded from the final set on business-communication grounds, not statistical grounds, and replaced by ELEVATORS_AVG.

Correlation-based reduction: 30 → 22 variables

Sign-check on the initial 30-variable logistic regression flagged 8 of 30 coefficients with an unexpected sign (WoE convention: coefficient should be negative, since higher WoE = safer bin = lower log-odds of default). Systematic pairwise correlation scan (|r| ≥ 0.7) identified two redundant clusters:

Cluster	Variables (highest |r|)	Kept (highest IV)
Property attributes	TOTALAREA_MODE, LIVINGAREA_AVG, FLOORSMAX_AVG, ELEVATORS_AVG (r = 0.84–0.94)	FLOORSMAX_AVG
Bureau credit dates	BUREAU_DAYS_CREDIT_MEAN/MIN, _UPDATE_MEAN, _ENDDATE_MEAN, DAYS_ENDDATE_FACT_MEAN (r = 0.71–0.89)	BUREAU_DAYS_CREDIT_MEAN
Bureau amounts	BUREAU_AMT_CREDIT_SUM_DEBT_MEAN/SUM (r = 0.85); LIMIT_MEAN/SUM (r = 0.81)	The _MEAN variant of each pair

Rule: within each correlated pair, keep the higher-IV variable, drop the other. Result: 8 variables removed, 22 remain.

0.7417

AUC before

0.7408

AUC after

8 → 4

Sign inconsistencies

GROUPED STEPWISE DIAGNOSIS

Siddiqi's information-type grouping applied to the remaining 4 sign flips

4 variables still showed unexpected coefficient signs after correlation pruning. Siddiqi's grouped-regression method (Ch. 10, Exhibit 10.14) evaluates variables by information-type group before combining, deliberately entering the dominant group last:

GROUPS (IN ENTRY ORDER): personal/demographic (10) → current credit (2) → bureau history (7) → external score (3, entered last)

Cumulative step	AGE_YEARS	BUREAU_CREDIT_ACTIVE_CLOSED_COUNT
+ demographic	-0.4663 (correct)	—
+ current credit	-0.4323 (correct)	—
+ bureau history	-0.3068 (correct)	+0.4029 (unexpected)
+ external score	+0.1241 (flips)	+0.0247 (n.s., p = 0.63)

ABLATION CONFIRMS DOMINANCE

Model refit without any EXT_SOURCE variable: AGE_YEARS returns to -0.3202 (correct sign). AUC drops to 0.6845 (from 0.7408), quantifying how much predictive power the external score alone contributes.

AGE_YEARS's sign flip is explained by dominance from EXT_SOURCE. BUREAU_CREDIT_ACTIVE_CLOSED_COUNT is not, it is unexpected before EXT_SOURCE even enters, and becomes statistically non-significant (p = 0.63) in the full model. This diagnosis explains 1 of 4 remaining flips; the other 3 (including BUREAU_CREDIT_ACTIVE_CLOSED_COUNT) required a different approach, covered next.

FINAL PRUNING: BACKWARD ELIMINATION

From 22 to 17 variables, method named explicitly

Correlation and VIF ruled out multicollinearity as the cause of the remaining sign flips (BUREAU_DAYS_CREDIT_UPDATE_MAX, NAME_INCOME_TYPE_BINNED, BUREAU_CREDIT_ACTIVE_CLOSED_COUNT): all VIF values in the bureau cluster stayed below 2.2, well under the conventional concern threshold (5). With no isolable statistical cause, the remaining decision criterion was empirical: does removing the variable measurably reduce AUC?

Bureau-group variable	Max pairwise |r|	VIF
BUREAU_DAYS_CREDIT_ENDDATE_MEAN	0.63	2.14
BUREAU_DAYS_CREDIT_UPDATE_MAX	0.60	2.02
BUREAU_DAYS_CREDIT_MEAN	0.63	1.96
BUREAU_CREDIT_ACTIVE_ACTIVE_COUNT	0.59	1.82
BUREAU_AMT_CREDIT_SUM_DEBT_MEAN	0.60	1.76
BUREAU_CREDIT_ACTIVE_CLOSED_COUNT	0.53	1.50
BUREAU_AMT_CREDIT_SUM_LIMIT_MEAN	0.36	1.20

VIF rules out combined multicollinearity as the cause; it does not identify what the cause is. This is a valid, honest outcome, not a dead end: it tells us the remaining decision has to be made on predictive contribution, not on a diagnosed statistical defect.

BACKWARD ELIMINATION, STEP BY STEP

Step	n vars	AUC
Correlation pruning	22	0.7408
− BUREAU_CREDIT_ACTIVE_CLOSED_COUNT (p = 0.63, formal significance test)	21	0.7408
− AGE_YEARS, NAME_INCOME_TYPE_BINNED, BUREAU_DAYS_CREDIT_UPDATE_MAX (tested together)	18	0.7406
− BUREAU_DAYS_CREDIT_ENDDATE_MEAN (coef ≈ 0.011)	17	0.7406

Every step above is the same criterion applied twice, once formally (p-value, a statistical test of whether a coefficient could be zero given the other variables), once by direct ablation (remove and re-measure AUC). Total cost: 0.0011 AUC (0.7417 ⟶ 0.7406) to eliminate every sign inconsistency across 13 removed variables.

FINAL MODEL

17-variable logistic regression, all coefficients sign-consistent

Variable	Coefficient
EXT_SOURCE_2	-0.7628
EXT_SOURCE_3	-0.7553
NAME_EDUCATION_TYPE_BINNED	-0.5863
EXT_SOURCE_1	-0.5264
AMT_CREDIT	-0.5188
CODE_GENDER	-0.4689
YEARS_EMPLOYED	-0.3594
ORGANIZATION_TYPE_BINNED	-0.3495

Variable	Coefficient
FLOORSMAX_AVG	-0.3235
BUREAU_CREDIT_ACTIVE_ACTIVE_COUNT	-0.2549
BUREAU_AMT_CREDIT_SUM_LIMIT_MEAN	-0.2411
OCCUPATION_TYPE_BINNED	-0.2383
REGION_POPULATION_RELATIVE	-0.2029
DAYS_ID_PUBLISH	-0.1957
BUREAU_AMT_CREDIT_SUM_DEBT_MEAN	-0.1653
DAYS_LAST_PHONE_CHANGE	-0.1610
BUREAU_DAYS_CREDIT_MEAN	-0.0793

All 17 coefficients negative, consistent with WoE convention (higher WoE = safer bin = lower log-odds of default). No exceptions.

POINT SCALE CALIBRATION

Factor/Offset derivation and two calibration attempts

$$\text{Score} = \text{Offset} + \text{Factor} \times \ln(\text{odds})$$

$$\text{Factor} = \text{PDO} / \ln(2)$$

$$\text{Offset} = \text{Score\_ref} - \text{Factor} \times \ln(\text{Odds\_ref})$$

Attempt	Parameters	Result
1	PDO=20, Score=600, Odds=50:1 (market convention)	Range too narrow (402–582); anchor point (50:1) far safer than the actual 11.4:1 base odds
2	PDO=20, Score=500, Odds=11.4:1 (bad-rate-implied)	Mean did not land on 500 (439.6); class_weight='balanced' shifted the model's own intercept away from the true prior
Final	PDO=40, Score=500, Offset anchored on the model's own empirical mean log-odds on training data	Mean 500.16, std 52.72, range 312–669 (validation)

Key lesson: class_weight='balanced' recalibrates the model's intercept to treat classes as equally weighted, so the population's average predicted log-odds no longer matches the true 8.07% bad rate directly. The offset must anchor on the model's own empirical average, not a theoretical bad-rate-implied value.

Point contribution per bin: base_points_per_characteristic + (−Factor × coefficient × WoE), with the constant term (Offset − Factor × intercept) distributed evenly across the 17 characteristics. Validated additively: summed points reproduce the model's direct score within rounding.

VALIDATION

AUC, KS, Gini, and leakage controls

AUC	KS	Gini
0.7406	0.3621	0.4812

Gini = 2 × AUC − 1 (0.4812, confirmed). KS in the "reasonable to good" range by common thresholds (30–40).

LEAKAGE CONTROLS

- WoE tables and bin edges frozen strictly on the training split; validation data is only ever looked up, never used to recompute bins or categories
- Numeric bins matched by position (pd.cut with fixed edges, labels=False), not by Interval-object identity, avoiding a floating-point mismatch bug found and fixed during development
- Categorical rare-category collapsing frozen as an explicit raw-category-to-bucket dictionary; unseen categories at scoring time fall back to 'Other_grouped' or 'Missing', never a new bucket

Absolute and relative default-rate reduction

Rejected	XGBoost (pp)	Scorecard (pp)	Scorecard (relative %)
5%	1.33	1.19	14.7%
10%	2.10	1.95	24.2%
15%	2.73	2.45	30.4%
20%	3.24	2.93	36.3%
25%	3.68	3.38	41.9%
30%	4.04	3.76	46.6%

~90–93%

of the XGBoost baseline's business gain is retained by the 17-variable scorecard, at every rejection threshold tested